

Day 4: Hands-on Subnetting

Task 1 — Divide 192.168.10.0/24 into 4 Subnets

Given network:

192.168.10.0/24

We need **4 equal subnets**.

To create 4 subnets, we borrow **2 host bits**:

/24 + 2 = /26

Subnet mask:

255.255.255.192

Each subnet has **64 total addresses** and **62 usable host addresses**.

Subnet	Network Address	Usable Host Range	Broadcast Address
Subnet 1	192.168.10.0/26	192.168.10.1 – 192.168.10.62	192.168.10.63
Subnet 2	192.168.10.64/26	192.168.10.65 – 192.168.10.126	192.168.10.127
Subnet 3	192.168.10.128/26	192.168.10.129 – 192.168.10.190	192.168.10.191
Subnet 4	192.168.10.192/26	192.168.10.193 – 192.168.10.254	192.168.10.255

Task 2 — Assign IPs to 5 Devices

I selected **Subnet 1: 192.168.10.0/26**.

Usable IP range:

192.168.10.1 – 192.168.10.62

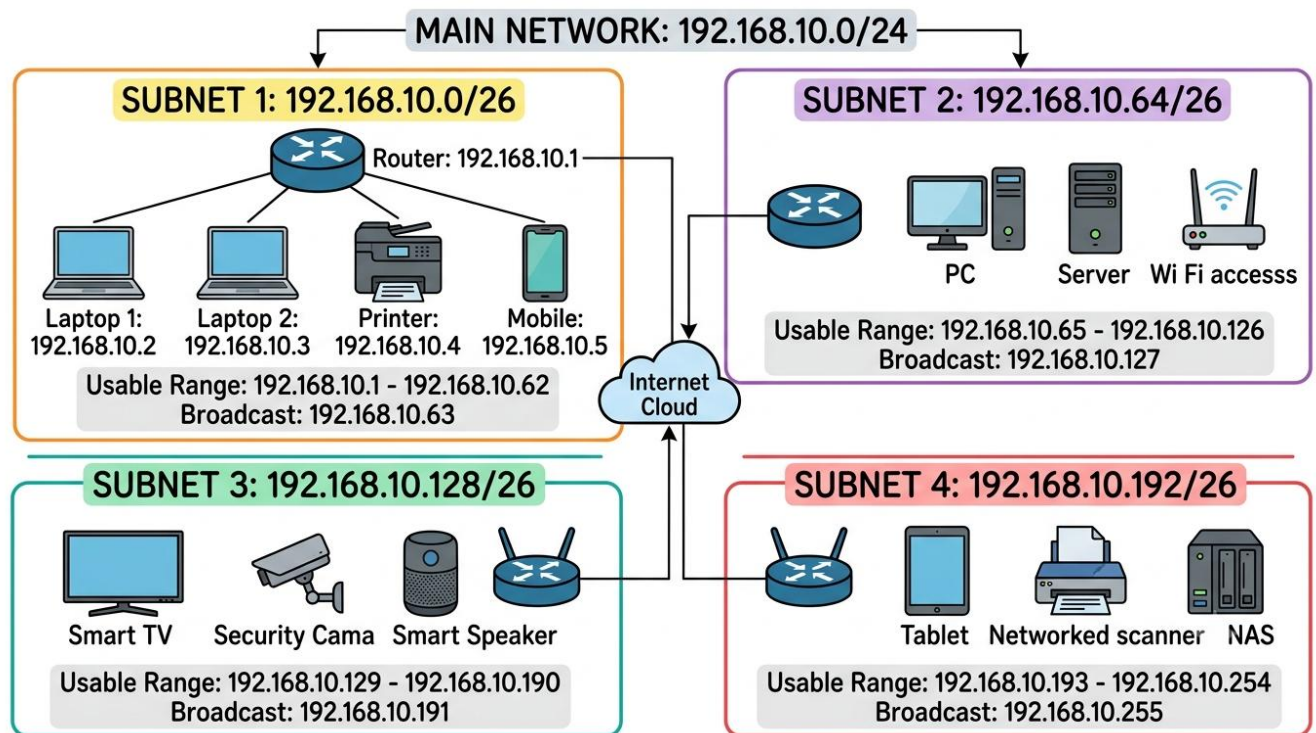
I assigned the following IP addresses:

Device	IP Address	Subnet Mask
Router	192.168.10.1	255.255.255.192
Laptop 1	192.168.10.2	255.255.255.192
Laptop 2	192.168.10.3	255.255.255.192
Printer	192.168.10.4	255.255.255.192
Mobile	192.168.10.5	255.255.255.192

Default Gateway: 192.168.10.1

I did not assign the network address 192.168.10.0 or broadcast address 192.168.10.63 to any device.

Task 3 — Subnet Plan Diagram



The network **192.168.10.0/24** is divided into four equal /26 subnets. I assigned five devices to the first subnet using valid host IP addresses.

Task 4 — Step-by-Step Subnet Calculation

Step 1 — Start with the Network

The given network is:

192.168.10.0/24

A /24 network has 256 total IP addresses.

Step 2 — Need 4 Equal Subnets

To create 4 subnets:

$$2^2 = 4$$

So, I borrow **2 bits** from the host portion.

$$/24 + 2 = /26$$

Step 3 — Find the Subnet Mask

The /26 subnet mask is:

255.255.255.192

Step 4 — Find Addresses Per Subnet

There are 6 host bits remaining:

$$32 - 26 = 6$$

Total addresses:

$$2^6 = 64$$

Usable host addresses:

$$64 - 2 = 62$$

Two addresses are reserved:

- . Network address
- . Broadcast address

Step 5 — Find the Block Size

The block size is:

$$256 - 192 = 64$$

So the subnet addresses increase by 64:

$$0 \rightarrow 64 \rightarrow 128 \rightarrow 192$$

Step 6 — Find the Four Subnets

The four network addresses are:

- . 192.168.10.0
- . 192.168.10.64

- 192.168.10.128
- 192.168.10.192

Step 7 — Assign Device IPs

I selected the first subnet and used its valid host range:

192.168.10.1 – 192.168.10.62

Then I assigned:

- Router → 192.168.10.1
- Laptop 1 → 192.168.10.2
- Laptop 2 → 192.168.10.3
- Printer → 192.168.10.4
- Mobile → 192.168.10.5

Result:

The /24 network was successfully divided into four equal /26 subnets, and valid IP addresses were assigned to five devices without using the network or broadcast addresses.